

T-Shirt Folding SOP (Men's or Women's)

This SOP covers folding a single short-sleeve t-shirt (men's or women's) into a clean folded bundle using a two-arm robot system. Both arms are used throughout: the grippers cooperate to lift, straighten, tension, and fold the t-shirt, with the left and right grippers taking the specific roles called out in each step. The task runs from a crumpled t-shirt at the back-left of the table to a folded t-shirt (or folded stack) at the back-right.

What stays constant across all sessions:

- **Working position:** the t-shirt is always moved to the centre of the table before straightening and folding.
- **Straighten target:** every t-shirt must lie flat, **face down, with the collar on the right** before any folding begins.
- **Fold order:** the near side is folded first, then the far side, then the collar end is folded down to meet the hem to complete the fold.

Setup

Go through both checklists before starting the episode.

Hardware checklist

- ☐ Cameras are on and recording
- ☐ Env camera frame includes the front and back edges of the table and is centered on the table's midpoint
- ☐ Both arms are at the home position with grippers open
- ☐ Table surface is clear of any objects other than the t-shirt(s)

Materials checklist

- ☐ **T-shirt(s)** to be folded are crumpled and placed at the **back-left** of the table
- ☐ **Center** of the table is clear (working area for straightening and folding)
- ☐ **Back-right** of the table is clear (output location for the folded t-shirt / stack)

Workspace layout

- **Back-left:** crumpled t-shirt(s) pile (input)
- **Center:** working area (straighten + fold)
- **Back-right:** folded t-shirt / folded stack (output)

Vocabulary

These are the terms used in this SOP. Operators and annotators must use this language consistently. One term per concept, used throughout.

T-shirt anatomy

- **T-shirt:** a single short-sleeve garment to be folded.
- **Collar:** the neck opening of the t-shirt.
- **Bottom hem:** the bottom edge of the t-shirt body, opposite the collar.
- **Sleeve:** one of the two short arms of the t-shirt.
- **Sleeve corner:** the outer bottom corner of a sleeve cuff.
- **Bottom corner:** one of the two corners where the bottom hem meets a side hem.
- **Side hem:** the side edge of the body running between a sleeve corner and a bottom corner.
- **Face down:** the front of the t-shirt rests against the table.
- **Face up:** the front of the t-shirt faces the camera.
- **Near side:** the side of the t-shirt closest to the operator/front of the table.
- **Far side:** the side of the t-shirt farthest from the operator/back of the table.
- **Folded bundle:** the finished square-ish result after the main fold.

Workspace zones

- **Back-left edge:** input zone holding crumpled t-shirts at the back-left of the table.
- **Working area:** the center of the table where straightening and folding happen.
- **Back-right edge:** output zone where the folded t-shirt/stack is placed at the back-right of the table.
- **Home position:** the default resting pose for each arm: gripper open and clear of the table.

Actions

- **Pick:** move one crumpled t-shirt to the working area.
- **Straighten:** lift, tension, and fling/drape the t-shirt so it lies flat, face down, collar on the right.
- **Fold:** fold a side or end of the t-shirt inward along a crease until it lies flat.
- **Stack:** place the newly folded t-shirt onto an existing pile (batch sessions only).

Steps

Step 1: Move a t-shirt to the working area

Goal: one t-shirt sits in the working area, ready to be straightened.

- With the left gripper, grasp one t-shirt from the back-left pile.
- Move it to the center of the table.

Step 2: Straighten the t-shirt

Goal: the t-shirt lies flat and face down, with the collar on the right.

2.1 Lift the t-shirt

- With the left gripper, grasp the highest point of the crumpled t-shirt and raise it until the bottom hem clears the table (or until full reach/extension).

2.2 Transfer grip (transfer to the right gripper)

- With the right gripper, grasp the lowest visible corner or sleeve, then release the left gripper.
- Raise the right gripper until the t-shirt hangs clear of the table (or the right arm is fully extended).

2.3 Find the adjacent corner (trace along the hem)

- With the left gripper, trace along the side hem from the current grasp point until you reach the next corner:
 - If holding a sleeve corner → trace to the nearest bottom corner.
 - If holding a bottom corner → trace to the nearest sleeve corner.

2.4 Tension the t-shirt

- Bring both grippers to the same height and pull outward in opposite directions to create tension along the hem.

2.5 Fling the t-shirt flat

- Fling the t-shirt so that both arms move towards the operator(front of the table) and the bottom of the t-shirt projects toward one side of the table and the collar is pulled toward the opposite side(left or right).
- If the t-shirt lands flat enough to identify orientation → continue. If it lands crumpled and orientation is unclear → return to 2.1.

2.6 Check orientation - front vs. back

- If face down → continue to 2.7.
- If face up → flip by draping (not flinging): confirm both near corners are accessible (un-tuck if needed), grasp the farthest sleeve corner and farthest bottom corner, lift the far

edge until the near edge peels off the table, bring the hanging edge towards operator and down, then move back and down in one smooth motion to drape the gripped side over so it lands behind the contact line, and release. If it lands crumpled → return to 2.1.

2.7 Check orientation - collar position

- If the collar is on the left → rotate the t-shirt 180° clockwise.
- If the collar is on the right → continue to 2.8.

2.8 Un-tuck any folded edges

- Inspect the four corners (two bottoms + two sleeves) and the collar. For any corner/collar tucked under: fold it inward toward the center, then unfold back outward, stretching the corner outward. Repeat until all corners and the collar lie flat.

Step 3: Fold the near side toward the center

Goal: the near side is folded in, with the resulting crease aligned to the near edge of the collar.

- With the right gripper, grasp the outer corner of the near sleeve (farthest from the collar, on the near side).
- With the left gripper, grasp the near bottom corner (the bottom-hem corner closest to the operator).
- Fold inward so the resulting horizontal crease aligns with the near edge of the collar, then make small adjustments to straighten the folded portion.

Step 4: Fold the far side toward the center

Goal: the far side is folded in, with the new crease parallel to the Step 3 crease and aligned to the far edge of the collar.

- With the right gripper, grasp the outer corner of the far sleeve.
- With the left gripper, grasp the far bottom corner.
- Fold inward so the resulting horizontal crease aligns with the far edge of the collar and parallel to the crease from Step 3, then straighten as needed.

Step 5: Reposition the folded shirt diagonally for the main fold

Goal: the t-shirt is set up diagonally so the collar end can be folded down to the hem.

- Move the folded t-shirt so the collar sits under the right-side working position (front-right area), the bottom hem points toward the back-left area, and the long axis is at roughly a 45° angle to the table's front edge (collar front-right → hem back-left).
- Multiple small grasp/release moves may be needed to rotate/translate. Keep the t-shirt flat and preserve the two parallel creases from Steps 3-4. If a crease opens, refold that side before continuing.

Step 6: Fold the collar end down to meet the hem

Goal: the t-shirt is folded roughly in half, with the collar meeting the hem in a square-ish bundle.

- With the left gripper, grasp the near collar-side corner, and with the right gripper, grasp the far collar-side corner.
- Lift and carry the collar end up and across toward the bottom hem at the center of the table.
- Land each collar corner directly on top of its corresponding hem corner (target: within ~3 cm).
- If crumpled after folding, grasp the back-right and front-left corners and pull in opposite directions to create tension; repeat for the other back-left and front-right corners.
- Release both grippers.

Warning: The shirt should now be folded roughly in half (collar meeting hem), producing a square-ish folded bundle.

Step 7: Stack onto a pile (batch sessions only)

Goal: the newly folded t-shirt is stacked squarely on the existing pile.

- If this is a single-t-shirt episode or the back-right pile is empty→ skip to Step 8.
- If stacking onto an existing pile:
 - With the left gripper, grasp the middle of the left edge of the newly folded shirt and pull it down to the left to clear stacking space.
 - With the right gripper, grasp the existing pile by the middle of its left edge and drag it down next to the new folded t-shirt (within 10-20 cm, aligned front-to-back).
 - With the left gripper grasp the top-left corner and with the right gripper grasp the bottom-right corner; lift the new folded t-shirt and place it on top (within 3 cm tolerance) so the fold edges align at the front (closed edge facing front) and the new t-shirt is centered side-to-side over the t-shirt below (midpoints aligned).
 - Release both grippers.

Step 8: Move the folded t-shirt or stack to the back-right edge

Goal: the folded t-shirt/stack rests in the designated back-right output zone.

- With the right gripper, grasp the folded t-shirt/stack along the middle of its right edge.
- Drag it back and to the right into the output location at the back-right of the table.
- Release the gripper.

Step 9: Return to home and end the episode

- Move both arms back to the home position with grippers open.
- End data collection.

After each episode: repeat or reset

- **Batch sessions (e.g., 5x):** if fewer than the target number are folded, return to Step 1 with the next t-shirt. Once the target number is folded and stacked, reset the workspace before the next session.
- Clear the working area and re-run the Setup checklist before the next session.

SOP violations

These are the things the operator can do that break (violate) the SOP. This list feeds the violation set and is what reviewers look for using the side-by-side review tool.

How to record a violation in review

For each violation you spot in a recorded episode, record:

1. The **start timestamp** of the violation in the video.
2. The **violation name** from the list below.
3. The **SOP rule broken** (the step number from the list below).

The **visible cue** is what you actually see in the video. The **coaching note** is for retraining the operator after the review; it is not what the annotator labels.

Critical violations (Rate the episode)

A Critical violation means the episode produced data that would teach the policy bad behavior. The episode must be flagged with less ratings and must not be used for training.

Critical: T-shirt dropped off the table or out of the working area.

- **Visible cue:** during any handling (Steps 1, 2, 5, 6, 8) the t-shirt separates from the gripper and falls off the table or lands outside the working area.
- **SOP rule broken:** Steps 1-8 (the t-shirt stays in controlled handling within the working area).
- **Coaching note:** grip too loose or release too early/high. Maintain a secure from-above grip and lower into position before releasing.

Critical: Final result not folded (crumpled or open bundle).

- **Visible cue:** at the end of Step 6, the t-shirt is not a clean folded bundle - it is crumpled, the main fold is incomplete, or a side has come unfolded.

- **SOP rule broken:** Step 6 goal (t-shirt folded roughly in half, collar meeting hem, square-ish bundle).
- **Coaching note:** creases not followed or corners not landed on the hem. Re-tension and refold per the SOP before proceeding.

Critical: Wrong fold order.

- **Visible cue:** the operator folds the far side before the near side, or performs the collar-to-hem main fold (Step 6) before both side folds (Steps 3-4) are complete.
- **SOP rule broken:** Step 3 must precede Step 4, and Steps 3-4 must precede Step 6.
- **Coaching note:** out-of-order folds prevent clean, parallel creases. Fold near side, then far side, then the collar ends over the hem.

Critical: Folded with wrong orientation (face-up or collar not on the right).

- **Visible cue:** folding begins (Step 3) while the t-shirt is face-up, or with the collar on the left rather than the right.
- **SOP rule broken:** Step 2 goal (lies flat, face down, collar on the right) before any folding.
- **Coaching note:** complete the orientation checks in 2.6-2.7 before folding. Flip by draping and/or rotate 180° clockwise as needed.

Critical: Sleeve or edge protrudes from the finished fold.

- **Visible cue:** at the end of Step 6, a sleeve, corner, or loose edge sticks out of the folded bundle.
- **SOP rule broken:** Step 4 (side fully tucked) and Step 6 (clean bundle).
- **Coaching note:** ensure the Step 4 fold is fully tucked before Step 6. Re-tuck the protruding edge and refold.

Critical: Wrong arm used for an action.

- **Visible cue:** any step that specifies the left gripper or the right gripper is performed with the opposite gripper.
- **SOP rule broken:** any step that specifies a gripper, including Steps 1, 2.1, 2.2, 3, 4, 6, 7, and 8.
- **Coaching note:** operator confusion about left vs right roles. Walk through the SOP step by step with the operator.

Critical: Stack misaligned or toppled (batch sessions).

- **Visible cue:** in Step 7 the new t-shirt is placed off the pile, the pile topples, or t-shirts slide apart.
- **SOP rule broken:** Step 7 (place centered, fold edges aligned, within tolerance).
- **Coaching note:** square the pile and folded shirt before placing, and use the specified corner grasps.

Critical: Setup checklist not satisfied at episode start.

- **Visible cue:** in the first second of the recorded video, any of the following is true: cameras not recording, env camera frame does not include front/back table edges or is not centered on the table's midpoint, either arm not at home position with gripper open, table surface holds objects other than the t-shirt(s), or materials checklist conditions (crumpled t-shirt(s) at back-left, center clear, back-right clear) are not met.
- **SOP rule broken:** the Setup section (Hardware and Materials checklists must both pass before recording starts).
- **Coaching note:** the operator skipped the checklist. Make running the checklist explicit before pressing record.

Minor violations (flag in review)

A Minor violation should be tagged with the timestamp and name; the episode handling depends on the data-pipeline rule below.

Minor: Re-fling or re-drape needed once.

- **Visible cue:** the t-shirt lands crumpled or face-up and the operator repeats the fling/drape (2.5/2.6) exactly once before proceeding.
- **SOP rule broken:** Step 2 (straighten flat, face down, in one fling/drape).
- **Coaching note:** a single re-attempt is tolerated but flagged. Aim for one smooth fling/drape close to the front.

Minor: Single re-grip on a pick or grasp.

- **Visible cue:** the operator closes on the t-shirt, finds the grip off, opens, and re-grips exactly once before lifting.
- **SOP rule broken:** Steps 1/2.1/2.2 (grasp securely, e.g., the highest point or correct corner).
- **Coaching note:** approach angle off. Practice the from-above approach so the first grip catches the right point.

Minor: Crease opened and refolded once.

- **Visible cue:** a Step 3 or Step 4 crease opens during repositioning and the operator refolds that side exactly once.
- **SOP rule broken:** Step 5 (preserve the two parallel creases).
- **Coaching note:** a single refold is tolerated but flagged. Keep the t-shirt flat while repositioning.

Minor: Collar corner lands off the hem corner (within tolerance).

- **Visible cue:** in Step 6 a collar corner lands more than a clean distance from its hem corner but still within ~3 cm, with no crumpling.
- **SOP rule broken:** Step 6 (land each collar corner on its hem corner, within ~3 cm).
- **Coaching note:** aim the collar corners more precisely before releasing.

Minor: Repeated fiddling with placement or folds.

- **Visible cue:** the operator makes many small adjustments (more than about two) to settle the t-shirt, a side, or the bundle, rather than settling it in one or two corrections.
- **SOP rule broken:** the straighten/fold sub-steps (settle in one or two small adjustments).
- **Coaching note:** grip or approach is off, forcing repeated correction. Tighten the approach so each fold lands close to target.

Minor: Output placement at back-right slightly off zone.

- **Visible cue:** in Step 8 the folded t-shirt/stack is released at the back-right but ends up noticeably off from the designated zone, without falling or requiring further correction.
- **SOP rule broken:** Step 8 (move the folded t-shirt/stack into the back-right output zone).
- **Coaching note:** aim the release point more precisely; push the bundle deep into the back-right corner.

Minor: Jerky or stepped carry or fold.

- **Visible cue:** a carry, fold, or drag in Steps 1, 2, 5, 6, or 8 is not one smooth motion; visible pauses, jerks, or back-and-forth.
- **SOP rule broken:** the carry/fold sub-steps (smooth, controlled motion).
- **Coaching note:** practice continuous motion; smooth out carries and folds, keeping the t-shirt closer to the front.

Minor: Stack slightly off-center but within tolerance.

- **Visible cue:** in Step 7 the new t-shirt lands within the 3 cm tolerance but is visibly off-center side-to-side or at the front edge.
- **SOP rule broken:** Step 7 (centered side-to-side, fold edges aligned at the front).
- **Coaching note:** square both pieces and aim the corner grasps before placing.

Minor: Arms not fully at home position at episode end.

- **Visible cue:** in the final frame, both arms are close to the home position but not exactly at it (gripper not fully open, or arm position visibly off home).
- **SOP rule broken:** Step 9 (return to home position with grippers open).
- **Coaching note:** complete the home motion explicitly before ending recording.

Minor-episode handling rule

Open question (resolve before locking of the violation set): how should episodes with only Minor violations (no Critical violations) be handled in the data pipeline? Pick one of the following and document it here:

- **Option A:** include the episode in training data, tagged with the Minor violation name.
- **Option B:** exclude the episode from training data; use it only for evaluation.
- **Option C:** hold the episode out for case-by-case review by the coordinator before assignment.

Default until decided: **Option A** (include with tag). This matches the goal of capturing realistic operator variance in training.

Non-violation failures

These are episode failures that are not the operator's fault and do not go in the violation set. They are recorded as system issues, the episode is discarded/deleted, and the episode does not become a coaching point for the operator.

- **Recording stopped or paused mid-episode:** Cause: software or hardware issue with the recording system.
- **Camera dropped frames or lost feed during the episode:** Cause: camera or capture system issue.
- **Hardware fault on the robot arm:** Cause: gripper malfunction, arm position drift, or motor error during the episode.
- **Defective t-shirt:** Cause: a torn, stained, or misshapen t-shirt that cannot be folded cleanly, through no fault of the operator. Replace before the next episode.

After the episode: reset the workspace

This part is not recorded. It is just how you reset the table for the next episode.

- With recording off, move the folded shirt(s) from the back-right back to the back-left.
- Crumple each shirt individually so the back-left once again contains distinct crumpled shirts matching the initial setup state.
- Confirm the table surface is clear of any objects other than the t-shirt(s) in their designated zone.
- Go through the Setup checklist again before starting the next episode.

Warning: If a t-shirt fell to the table or floor during the episode, retrieve it and return it to the input zone before re-running setup. The table must be clear except for the t-shirt(s) at the back-left.

Annotation subtasks

1. Move one t-shirt from the input area to the center working area
2. Straighten the t-shirt (face down, collar on the right, arms out)
3. Fold the near sleeve and side toward the center
4. Fold the far sleeve and side toward the center
5. Reposition the t-shirt to a diagonal setup for the main fold
6. Fold the collar-side corners down to meet the bottom corners (close the fold)
7. Stack the folded t-shirt onto the pile (batch only)
8. Move the folded t-shirt or stack to the back-right output area
9. Home the arms